Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

The equation $(\cos a - 1)x^2 + (\cos a)x + \sin a = 0$ in x has real roots, then possible set of values of x may be

(A) $[0, 2\pi]$

(B) $[-\pi, 0]$

(C) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

(D) $[0, \pi]$

Q.2

If $\sin\theta + \sin 3\theta + \sin 5\theta = 0$, then the general value of θ is -

(A) $\frac{n\pi}{6}, \frac{m\pi}{12}; m, n \in I$

(B) $\frac{n\pi}{3}, \ m\pi \pm \frac{\pi}{3}; m, n \in I$

(C) $\frac{n\pi}{3}, \ m\pi \pm \frac{\pi}{6}; m, n \in I$

(D) None of these

Q.3

If $2\cos^2\theta + 3\sin\theta = 0$, then general value of θ is -

(A) $n\pi + (-1)^n \frac{\pi}{6}; n \in I$

(B) $2n\pi \pm \frac{\pi}{6}; n \in I$

(C) $n\pi + (-1)^{n+1} \frac{\pi}{6}; n \in I$

(D) None of these

Q.4

If $\lambda\cos 2\theta = \cos(\theta + \alpha), \lambda\sin 2\theta = 2\sin(\theta + \alpha)$then $\cos^{2/3}\alpha + \sin^{2/3}\alpha =$

(A) λ

(B) $\lambda^{1/3}$

(C) $\lambda^{2/3}$

(D) $\lambda^{5/3}$

Q.5

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If $\sin^3 x \sin 3x = \sum_{r=0}^{n} c_r \cos rx$ where $C_0, C_1, C_2 \ldots\ldots\ldots\ldots C_n$ are constants, $C_n \neq 0$ then $n =$

(A) 2

(B) 4

(C) 6

(D) 8

Q.6

The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to

(A) $3-\sqrt{3}$

(B) $2(3-\sqrt{3})$

(C) $2(\sqrt{3}-1)$

(D) $2(2+\sqrt{3})$

Q.7

Which of the following is not equal to 1 ?

(A) $\cos^4\theta - \sin^4\theta + 2\sin^2\theta$

(B) $\frac{\sin^2\theta}{2}\left(1+\cot^2\theta\right) + \frac{\cos^2\theta}{2}\left(1+\tan^2\theta\right)$

(C) $\sin^2\theta\cos^2\phi + \cos^2\theta\sin^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta\cos^2\phi$

(D) $\sin^4\theta - \cos^4\theta - 2\sin^2\theta$

Q.8

Solve : $\tan\theta + \tan(\theta+(\pi/3)) + \tan(\theta+(2\pi/3)) = 3$

(A) $(3n+1)(\pi/12)$, where $n \in I$

(B) $(4n+1)(\pi/6)$, where $n \in I$

(C) $(4n+1)(\pi/12)$, where $n \in I$

(D) None of these

Q.9

In which of the following sets the inequality $\sin^6 x + \cos^6 x > \frac{5}{8}$ holds good ?

(A) $\left(-\frac{\pi}{8}, \frac{\pi}{8}\right)$

(B) $\left(\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

(C) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

(D) $\left(\frac{7\pi}{8}, \frac{9\pi}{8}\right)$

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Q.10

For the system of equations, $\sin x + \cos y = 1$ and $\cos 2x - \cos 2y = 1$. The solutions having same sign for sine, and cosine for x and y respectively.

(A) all the solutions of given equation.

(B) $x = n\pi + (-1)^n \frac{\pi}{6}, y = 2n\pi \pm \frac{\pi}{3}$

(C) $x = 2n\pi + \frac{\pi}{7}, y = 2n\pi + \frac{2\pi}{7}$

(D) none of these

Q.11

For the smallest positive values of x and y the equation, $2(\sin x + \sin y) - 2\cos(x - y) = 3$, has a solution then which of the following is/are true -

(A) $\sin \frac{x+y}{2} = 1$

(B) $\cos\left(\frac{x-y}{2}\right) = \frac{1}{2}$

(C) Number of ordered pairs (x, y) is 2

(D) Number of ordered pairs (x, y) is 3

Q.12

Passage

If $\alpha, \beta, \gamma, \delta$ are the solutions of the equation $\tan\left(\theta + \frac{\pi}{4}\right) = 3\tan 3\theta$, no two of which have equal tangents. Then,

Question:

The value of $\tan\alpha + \tan\beta + \tan\gamma + \tan\delta$ is $-$

(A) $1/3$

(B) 8/3

(C) $-8/3$

(D) 0

Q.13

Passage

If $\alpha, \beta, \gamma, \delta$ are the solutions of the equation $\tan\left(\theta + \frac{\pi}{4}\right) = 3\tan 3\theta$, no two of which have equal tangents. Then,

Question:

The value of $\tan\alpha \tan\beta \tan\gamma \tan\delta$ is -

(A) $-1/3$

(B) -2

(C) 1

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(D) None of these

Q.14

Passage

If $\alpha, \beta, \gamma, \delta$ are the solutions of the equation $\tan\left(\theta+\frac{\pi}{4}\right)=3\tan 3\theta$, no two of which have equal tangents. Then,

Question:

The value of $\frac{1}{\tan\alpha}+\frac{1}{\tan\beta}+\frac{1}{\tan\gamma}+\frac{1}{\tan\delta}$ is $-$

(A) -8

(B) 8

(C) $2/3$

(D) $1/3$

Q.15

Passage

Consider the system of equations : $\sin x\cos 2y=\left(a^2-1\right)^2+1, \cos x\sin 2y=a+1$

Question:

Number of values of a for which the system has a solution is -

(A) 1

(B) 2

(C) 3

(D) Infinite

Q.16

Passage

Consider the system of equations : $\sin x\cos 2y=\left(a^2-1\right)^2+1, \cos x\sin 2y=a+1$

Question:

Number of values of $x \in [0, 2\pi]$ when the system has solution for permissible values of a -

(A) 1

(B) 2

(C) 3

(D) 4

Q.17

Passage

Consider the system of equations : $\sin x\cos 2y=\left(a^2-1\right)^2+1, \cos x\sin 2y=a+1$

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Question:

Number of values of $y \in [0, 2\pi]$ when the system has solution for permissible values of a -

(A) 2

(B) 3

(C) 4

(D) 5

Q.18

Match the column

Column I (Equation)	**Column II (Solution)**
(A) $\max_{\theta \in R} \{5 \sin \theta + 3 \sin (\theta - \alpha)\} = 7$ then the set of possible values of α is	(p) $x = 2n\pi + 3\pi/4, n \in Z$
(B) $x \neq \frac{n\pi}{2}$ and $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$	(q) $x = 2n\pi \pm \pi/3, n \in Z$
(C) $\sqrt{(\sin x)} + 2^{1/4} \cos x = 0$	(r) $x = 2n\pi + \cos^{-1}(1/3), n \in Z$
(D) $\log_5 \tan x = (\log_5^4)(\log_4^{(3 \sin x)})$	(s) no solution

(A) A-p ; B-p ; C-s ; D-r

(B) A-q ; B-s ; C-p ; D-r

(C) A-q ; B-p ; C-p ; D-p

(D) A-s ; B-q ; C-r ; D-q

Q.19

The number of solutions of the equation
$\sin 2x + \cos 2x + 4 \sin x = 1 + 4 \cos x$ being in
the interval $[-6, 6]$, is

Q.20

Let $f(x) = \left(\frac{a^{\sin^4 x}}{a^{2\sin^2 x}}\right)^3 \cdot \left(a^{5\sin^3 x} \cdot a^{5 \sin x}\right)$ and $b = (\sqrt{5})^{2\log_5 3} (a > 0, a \neq 1)$. If the minimum value of $\left(f(x) \cdot a^b\right)^{\text{cosec}^2 x} + \log_a \sin x$ is equal to $a^k, k \in N$, find k.

Answers & Solutions

Q.1 Answer:

$[0, \pi]$

Solution:

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For real $x, D \geq 0$

$\therefore \quad \cos^2 a - 4(\cos a - 1)\sin a \geq 0$

$\Rightarrow \quad \cos^2 a - 4\sin a \cos a + 4\sin a \geq 0 \Rightarrow (\cos a - 2\sin a)^2 + 4\sin a - 4\sin^2 a \geq 0$

$\Rightarrow \quad (\cos a - 2\sin a)^2 + 4\sin a(1 - \sin a) \geq 0 \quad \ldots (i)$

$(\cos a - 2\sin a)^2 \geq 0,$ and $(1 - \sin a) \geq 0$

$\therefore$ **Inequality (i) will be definitely true when** $0 \leq a \leq \pi$

Q.2 Answer:

$\frac{n\pi}{3},\ m\pi \pm \frac{\pi}{3}; m, n \in I$

Solution:

If $(\sin 5\theta + \sin\theta) + \sin 3\theta = 0$

or, $2\sin 3\theta \cos 2\theta + \sin 3\theta = 0$

or, $\sin 3\theta(2\cos 2\theta + 1) = 0$

Case I : $\sin 3\theta = 0 \Rightarrow 3\theta = n\pi; n \in I \Rightarrow \theta = \frac{n\pi}{3}; n \in I$

Case II : $2\cos 2\theta + 1 = 0 \Rightarrow \cos 2\theta = -\frac{1}{2} \Rightarrow \cos 2\theta = \cos\frac{2\pi}{3}$

$\Rightarrow \quad \theta = m\pi \pm \frac{\pi}{3}; m \in I$

So the general solution of the given equation is

$\theta = \frac{n\pi}{3}$ **and** $\theta = m\pi \pm \frac{\pi}{3}$, **where** $m, n \in I$

Q.3 Answer:

$n\pi + (-1)^{n+1}\frac{\pi}{6}; n \in I$

Solution:

If $2\cos^2\theta + 3\sin\theta = 0 \Rightarrow 2\left(1 - \sin^2\theta\right) + 3\sin\theta = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta - 2 = 0$

$\Rightarrow 2\sin^2\theta - 4\sin\theta + \sin\theta - 2 = 0$

$\Rightarrow \quad 2\sin\theta(\sin\theta - 2) + (\sin\theta - 2) = 0$

$\Rightarrow \quad (\sin\theta - 2)(2\sin\theta + 1) = 0$

Case I : If $\sin\theta - 2 = 0$

$\sin\theta = 2$

Which is not possible because $-1 \leq \sin\theta \leq 1$

Case II : If $2\sin\theta + 1 = 0$

$\Rightarrow \quad \sin\theta = -\frac{1}{2}$ **or,** $\sin\theta = \sin\left(\frac{-\pi}{6}\right) \Rightarrow \theta = n\pi + (-1)^n\left(\frac{-\pi}{6}\right); n \in I$

$\Rightarrow \quad \theta = n\pi + (-1)^{n+1}\left(\frac{\pi}{6}\right); n \in I$

Q.4 Answer:

$\lambda^{2/3}$

Solution:

$\cdot \tan 2\theta = 2\tan(\theta + \alpha)$

$\frac{2\tan\theta}{1-\tan^2\theta} = 2\left(\frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha}\right) \Rightarrow \tan\theta = (\tan\alpha)^{1/3}$

$\cos\theta = \frac{\cos^{1/3}\alpha}{\sqrt{\cos^{2/3}\alpha + \sin^{2/3}\alpha}}, \lambda = \frac{\cos(\theta+\alpha)}{\cos 2\theta}$

$= \cos\theta\cos\alpha(1 - \tan\theta\tan\alpha)\frac{\left(1+\tan^2\theta\right)}{\left(1-\tan^2\theta\right)}$

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$= \frac{\cos^{4/3}\alpha\left(1+\tan^{2/3}\alpha\right)^2}{\sqrt{\cos^{2/3}\alpha+\sin^{2/3}\alpha}} = \left(\cos^{2/3}\alpha+\sin^{2/3}\alpha\right)^{3/2}$

$\therefore \quad \lambda^{2/3} = \cos^{2/3}\alpha + \sin^{2/3}\alpha$

Q.5 Answer:

6

Solution:

$\sin 3x = 3\sin x - 4\sin^3 x \quad \therefore \sin^3 x = \left|\frac{3}{4}\sin x - \frac{1}{4}\sin 3x\right|$

$\sin^3 x \sin 3x = \left|\frac{3}{4}\sin x - \frac{1}{4}\sin 3x\right| \sin 3x$

$= \frac{3}{8}\cdot 2\sin x \sin 3x - \frac{2}{8}\sin^2 3x = \frac{3}{8}(\cos 2x - \cos 4x) - \frac{1}{8}(1-\cos 6x)$

$\therefore C_0 = \frac{-1}{8}, C_2 = \frac{3}{8}, C_4 = \frac{-3}{8}, C_6 = \frac{1}{8}$

$C_1 = C_3 = C_5 = 0, n = 6$

Q.6 Answer:

$2(\sqrt{3}-1)$

Solution:

Here, $\sum_{k=1}^{13} \frac{1}{\sin\left\{\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right\}\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$

Converting into differences, by multiplying and dividing by

$\sin\left[\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)-\left\{\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right\}\right] = \sin\left(\frac{\pi}{6}\right)$

$\therefore \sum_{k=1}^{13} \frac{2\sin\left[\left(\frac{\pi}{4}+k\frac{\pi}{6}\right)-\left\{\frac{\pi}{4}+(k-1)\frac{\pi}{6}\right\}\right]}{\sin\left\{\frac{\pi}{4}+(k-1)\frac{\pi}{6}\right\}\sin\left(\frac{\pi}{4}+k\frac{\pi}{6}\right)\}}$

$\therefore \sum_{k=1}^{13} 2\left[\cot\left\{\frac{\pi}{4}+(k-1)\frac{\pi}{6}\right\} - \cot\left(\frac{\pi}{4}+k\frac{\pi}{6}\right)\right]$

$= 2\left\{\cot\frac{\pi}{4} - \cot\left(\frac{\pi}{4}+13\frac{\pi}{6}\right)\right\}$

$= 2\left[1-\cot\left(\frac{29\pi}{12}\right)\right] = 2\left[1-\cot\left(2\pi+\frac{5\pi}{12}\right)\right]$

$= 2\left[1-\cot\frac{5\pi}{12}\right]$

$= 2(1-2+\sqrt{3})$

$= 2(\sqrt{3}-1)$

Q.7 Answer:

$\sin^4\theta - \cos^4\theta - 2\sin^2\theta$

Solution:

(a) $\cos^4\theta - \sin^4\theta + 2\sin^2\theta$

$\left(\cos^2\theta+\sin^2\theta\right)\left(\cos^2\theta-\sin^2\theta\right)+2\sin^2\theta = 1$

(b) $\frac{\sin^2\theta}{2}\left(1+\cot^2\theta\right) + \frac{\cos^2\theta}{2}\left(1+\tan^2\theta\right)$

$= \frac{\sin^2\theta\,\mathrm{cosec}^2\theta}{2} + \frac{\cos^2\theta\sec^2\theta}{2} = \frac{1}{2}+\frac{1}{2} = 1$

(c) $\left(\sin^2\theta+\cos^2\theta\right)\left(\sin^2\phi+\cos^2\phi\right) = 1\times 1 = 1$

(d) $\sin^4\theta - \cos^4\theta - 2\sin^2\theta = \left(\sin^2\theta+\cos^2\theta\right)\left(\sin^2\theta-\cos^2\theta\right) - 2\sin^2\theta = -1$

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Q.8 Answer:

$(4n+1)(\pi/12)$, where $n \in I$

Solution:

From the given equation,

$\tan\theta + \frac{\tan\theta+\tan(\pi/3)}{1-\tan\theta\tan(\pi/3)} + \frac{\tan\theta+\tan(2\pi/3)}{1-\tan\theta\tan(2\pi/3)} = 3$

$\tan\theta + \frac{\tan\theta+\sqrt{3}}{1-\sqrt{3}\tan\theta} + \frac{\tan\theta-\sqrt{3}}{1+\sqrt{3}\tan\theta} = 3 \quad$ or $\quad 3\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta} = 3$

or $\quad \tan 3\theta = 1 = \tan(\pi/4) \quad$ or $3\theta = n\pi + (\pi/4)$

or $\quad \theta = (4n+1)(\pi/12),$ where $n \in I$

Q.9 Answer:

$\left(-\frac{\pi}{8}, \frac{\pi}{8}\right)$

Solution:

$\left(\sin^2 x + \cos^2 x\right)^3 - 3\sin^2 x\cos^2 x\left(\sin^2 x + \cos^2 x\right) > \frac{5}{8}$

$\Rightarrow 1 - 3\sin^2 x\cos^2 x > \frac{5}{8}$

$\Rightarrow \frac{3}{8} > 3\sin^2 x\cos^2 x$

$\Rightarrow 1 - 2\sin^2 2x > 0 \Rightarrow \cos 4x > 0$

$\Rightarrow 4x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow 4x \in \left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right), n \in Z$

Q.10 Answer:

all the solutions of given equation.

Solution:

As equation (1) $\Rightarrow \quad \sin x + \cos y = 1$

$\Rightarrow \quad \cos 2x - \cos 2y = 1 \quad \Rightarrow \quad \sin^2 x + \cos^2 y = \dfrac{1}{2}$

$\Rightarrow \quad \sin x = \dfrac{1}{2}$ and equation (2) $\cos y = \dfrac{1}{2}$

All solutions are such that sine and cosine of x, y respectively are of same sign.

$\Rightarrow \quad x = n\pi + (-1)^n\frac{\pi}{6}, \quad y = 2n\pi \pm \frac{\pi}{3}$

Q.11 Answer:

$\sin\frac{x+y}{2} = 1$

Solution:

The given equation is $2(\sin x + \sin y) - 2\cos(x-y) = 3$

$\Rightarrow \quad 2 \times 2\sin\frac{x+y}{2}\cos\frac{x-y}{2} - 2\left[2\cos^2\frac{x-y}{2} - 1\right] = 3$

$\Rightarrow \quad 4\cos^2\left(\frac{x-y}{2}\right) - 4\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) + 1 = 0$

$\Rightarrow \quad \cos\left(\frac{x-y}{2}\right) = \frac{4\sin\left(\frac{x+y}{2}\right) \pm \sqrt{16\sin^2\left(\frac{x+y}{2}\right) - 16}}{8}$

$\therefore \quad \sin^2\left(\frac{x+y}{2}\right) \geq 1 \Rightarrow \sin\left(\frac{x+y}{2}\right) = \pm 1$

Since x and y are smallest and positive, $\sin\frac{x+y}{2} = 1$ **and** $\frac{x+y}{2} = \frac{\pi}{2}$ **i.e.,** $x + y = \pi$

Also, $\cos\left(\frac{x-y}{2}\right) = \frac{1}{2} \Rightarrow x - y = \frac{2\pi}{3}$ **or** $-\frac{2\pi}{3}$

From eq. (1) and (2), we get $\left(x = \frac{5\pi}{6}, y = \frac{\pi}{6}\right)$ **or** $\left(x = \frac{\pi}{6}, y = \frac{5\pi}{6}\right)$

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Q.12 Answer:

0

Solution:

We have $\tan\left(\theta+\frac{\pi}{4}\right)=3\tan 3\theta$

$\Rightarrow \quad \frac{1+\tan\theta}{1-\tan\theta}=3\times\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$

$\Rightarrow \quad \frac{1+t}{1-t}=3\left(\frac{3t-t^3}{1-3t^2}\right)$ (putting $t=\tan\theta$)

$\Rightarrow \quad 3t^4-6t^2+8t-1=0$

Hence, $S_1=$ sum of roots $=t_1+t_2+t_3+t_4=0$

$S_2=$ Sum of product of roots taken two at a time $=-2$

$S_3=$ Sum of product of roots taken three at time $=-8/3$

$S_4=$ Product of all roots $=-1/3$

Q.13 Answer:

$-1/3$

Solution:

We have $\tan\left(\theta+\frac{\pi}{4}\right)=3\tan 3\theta$,

$\Rightarrow \frac{1+\tan\theta}{1-\tan\theta}=3\times\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$

$\Rightarrow \frac{1+t}{1-t}=3\left(\frac{3t-t^3}{1-3t^2}\right)$ (putting $t=\tan\theta$)

$\Rightarrow \quad 3t^4-6t^2+8t-1=0$

Hence, $S_1=$ sum of roots $=t_1+t_2+t_3+t_4=0$

$S_2=$ Sum of product of roots taken two at a time $=-2$

$S_3=$ Sum of product of roots taken three at time $=-8/3$

$S_4=$ Product of all roots $=-1/3$

Q.14 Answer:

8

Solution:

We have $\tan\left(\theta+\frac{\pi}{4}\right)=3\tan 3\theta$,

$\Rightarrow \frac{1+\tan\theta}{1-\tan\theta}=3\times\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$

$\Rightarrow \frac{1+t}{1-t}=3\left(\frac{3t-t^3}{1-3t^2}\right)$ (putting $t=\tan\theta$)

$\Rightarrow \quad 3t^4-6t^2+8t-1=0$

Hence, $S_1=$ sum of roots $=t_1+t_2+t_3+t_4=0$

$S_2=$ Sum of product of roots taken two at a time $=-2$

$S_3=$ Sum of product of roots taken three at time $=-8/3$

$S_4=$ Product of all roots $=-1/3$

Q.15 Answer:

1

Solution:

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The given system is $\sin x \cos 2y = \left(a^2-1\right)^2 + 1,$

$\cos x \sin 2y = a + 1$ **.....(1)**

Since the LHS of the equations do not exceed 1, the given system may have solutions only for a's such that

$\left(a^2-1\right)^2 + 1 \leq 1$ **and** $-1 \leq a+1 \leq 1 \quad \ldots\ldots\ldots$ **(2)**

$\Rightarrow \quad \left(a^2-1\right)^2 + 1 \leq 1 \Rightarrow \left(a^2-1\right)^2 \leq 0$

$\Rightarrow \quad \left(a^2-1\right)^2 = 0 \Rightarrow a = 1$

But $a = 1$ **does not satisfy the** $\cos x \sin 2y = a + 1$

Thus $a = -1$ **only and then we get**

$\sin x \cos 2y = 1$

$\cos x \sin 2y = 0$

Adding and subtracting, we get

$\sin(x + 2y) = 1$

$\sin(x - 2y) = 1$

Obviously the system (4) yields

$x + 2y = 2n\pi + \frac{\pi}{2}$ **and** $x - 2y = 2m\pi + \frac{\pi}{2}; n, m \in I$

$\therefore \quad \mathrm{x} = (\mathrm{n}+\mathrm{m})\pi + \frac{\pi}{2}, \mathrm{y} = \frac{1}{2}(\mathrm{n}-\mathrm{m})\pi, \mathrm{n}, \mathrm{m} \in \mathrm{I}$

Q.16 Answer:

2

Solution:

The given system is $\sin x \cos 2y = \left(a^2-1\right)^2 + 1,$

$\cos x \sin 2y = a + 1$

(1)

Since the LHS of the equations do not exceed 1, the given system may have solutions only for a's such that

$\left(a^2-1\right)^2 + 1 \leq 1$ **and** $-1 \leq a+1 \leq 1 \quad \ldots\ldots\ldots$ **(2)**

$\Rightarrow \quad \left(a^2-1\right)^2 + 1 \leq 1 \Rightarrow \left(a^2-1\right)^2 \leq 0$

$\Rightarrow \quad \left(a^2-1\right)^2 = 0 \Rightarrow a = 1$

But $a = 1$ **does not satisfy the** $\cos x \sin 2y = a + 1$

Thus $a = -1$ **only and then we get**

$\sin x \cos 2y = 1$

$\cos x \sin 2y = 0$

Adding and subtracting, we get

$\sin(x + 2y) = 1$

$\sin(x - 2y) = 1$

Obviously the system (4) yields

$x + 2y = 2n\pi + \frac{\pi}{2}$ **and** $x - 2y = 2m\pi + \frac{\pi}{2}; n, m \in I$

$\therefore \quad \mathrm{x} = (\mathrm{n}+\mathrm{m})\pi + \frac{\pi}{2}, \mathrm{y} = \frac{1}{2}(\mathrm{n}-\mathrm{m})\pi, \mathrm{n}, \mathrm{m} \in \mathrm{I}$

Q.17 Answer:

5

Solution:

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The given system is $\sin x \cos 2y = (a^2-1)^2 + 1,$

$\cos x \sin 2y = a + 1$

(1)

Since the LHS of the equations do not exceed 1, the given system may have solutions only for a's such that

$(a^2-1)^2 + 1 \le 1$ **and** $-1 \le a+1 \le 1$ **..... (2)**

$\Rightarrow (a^2-1)^2 + 1 \le 1 \Rightarrow (a^2-1)^2 \le 0$

$\Rightarrow (a^2-1)^2 = 0 \Rightarrow a = 1$

But $a = 1$ **does not satisfy the** $\cos x$ **Thus** $a = -1$ **only and then we get**

$\sin x \cos 2y = 1$

$\cos x \sin 2y = 0$ **Adding and subtracting, we get**

(3)

$\sin(x+2y) = 1$

$\sin(x-2y) = 1$

Obviously the system (4) yields

$x + 2y = 2n\pi + \frac{\pi}{2}$ **and** $x - 2y = 2\,m\pi + \frac{\pi}{2}; n, m \in I$

$\therefore \quad x = (n+m)\pi + \frac{\pi}{2}, y = \frac{1}{2}(n-m)\pi, n, m \in I$

Q.18 Answer:

A-q ; B-s ; C-p ; D-r

Solution:

(A) q, (B) s, (C) p, (D) r

(A) $5\sin\theta + 3(\sin\theta\cos\alpha - \cos\theta\sin\alpha) = (5 + 3\cos\alpha)\sin\theta - 3\sin\alpha\cos\theta$

$\Rightarrow \quad \max_{\theta \in R}\{5\sin\theta + 3\sin(\theta - \alpha)\} = \sqrt{(5+3\cos\alpha)^2 + 9\sin^2\alpha} = \sqrt{34 + 30\cos\alpha}$

$\therefore$ **The given equation is** $34 + 30\cos\alpha = 49$

$\Rightarrow \quad \cos\alpha = \frac{1}{2} \Rightarrow \alpha = 2n\pi \pm \frac{\pi}{3}, n \in Z$

(B) $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1 \Rightarrow \sin^2 x - 3\sin x + 2 = 0$

$\Rightarrow \quad (\sin x - 2)(\sin x - 1) = 0 \Rightarrow \sin x = 1$

but this does not satisfy the equation becomes $0^\circ = 1$ **is absurd.**

(C) $\sqrt{(\sin x)} + 2^{1/4}\cos x = 0$ **(1)**

$\therefore \quad \sqrt{(\sin x)} > 0$ **and so** $\cos x < 0$

ie., x **lies in** 2nd **quadrant.**

Eq. (1) can be rewritten as $2^{1/4}\cos x = -\sqrt{(\sin x)}$

Squaring $\sqrt{2}\cos^2 x = \sin x$

$\Rightarrow \sqrt{2}\sin^2 x + \sin x - \sqrt{2} = 0 \Rightarrow (\sqrt{2}\sin x - 1)(\sin x + \sqrt{2}) = 0$

$\sin x \ne -\sqrt{2} \quad \therefore \sin x = 1/\sqrt{2}$**..... (2)**

From eq. (2) $\cos x < 0$ **so** $x = 3\pi/4 \in 2^{\text{nd}}$ **quadrant**

and so the general value of x **is given by** $x = 2n\pi + 3\pi/4, n \in Z$

(D) $\log_5 \tan x = \left(\log_5^4\right)\left(\log_4^{(3\sin x)}\right)$

$\Rightarrow \quad \log_5 \tan x = \log_5(3\sin x) \quad \ldots\ldots\ldots \quad (3)$

Since $\log x$ is real when $x > 0$

We have $\tan x > 0, \sin x > 0 \quad \therefore$

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Now equation (3) gives

$\tan x = 3 \sin x$ **or** $\cos x = 1/3$

Q.19 Answer:

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Solution:

Given equation $2 \sin x \cos x + 1 - 2 \sin^2 x + 4 \sin x = 1 + 4 \cos x$

$\rightarrow 2 \sin x (2 - \sin x) = 2 \cos x (2 - \sin x)$

$\Rightarrow (2 - \sin x)(\sin x - \cos x) = 0$

$\sin x = \cos x$ **or** $\tan x = 1$

$\rightarrow x = n\pi + \frac{\pi}{4}$

$x = \frac{-7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$

Q.20 Answer:

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Solution:

$\because f(x) = a^{3 \sin^4 x + 5 \sin^3 x - 6 \sin^2 x + 5 \sin x}$ and $b = 3 \quad (\sin x > 0)$

$\therefore \quad \left(f(x) \cdot a^3\right)^{\text{cosec}^2 x} = a^{3 \sin^2 x + 5 \sin x - 6 + \frac{5}{\sin x} + \frac{3}{\sin^2 x}}$

$= a^{3\left(\sin^2 x + \frac{1}{\sin^2 x}\right) + 5\left(\sin x + \frac{1}{\sin x}\right) - 6}$

$\because \quad$ min. value of $3\left(\sin^2 x + \dfrac{1}{\sin^2 x}\right) + 5\left(\sin x + \dfrac{1}{\sin x}\right) - 6$

$= 3(2) + 5(2) - 6 = 10 \quad (\because \sin x > 0$ and the value is minimum at $\sin x = 1)$

$\Rightarrow \log_a (\sin x) = 0$

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